THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL UNIT TEST SET 1

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Distance is a: 1

(a) vector **(b)** force **(c)** scalar **(d)** unit

2. Force is a: 1

(a) mass **(b)** speed **(c)** type of energy **(d)** push or pull

3. Work = 1

(a) force/displacement **(b)** mass x speed

(c) force x displacement **(d)** force + displacement

4. Sound is produced by: 1

(a) rest **(b)** heat **(c)** vibration **(d)** light

5. The area under a v-t graph gives: 1

(a) speed **(b)** acceleration **(c)** displacement **(d)** force

6. A heavier body has: 1

(a) more inertia **(b)** no inertia

(c) less inertia **(d)** same inertia as a lighter one

7. A fixed pulley: 1

(a) reduces the load **(b)** creates work

(c) changes the direction of force **(d)** has MA of 2

8. Ultrasonic sounds have frequency: 1

(a) above 20 kHz **(b)** 20 Hz to 20 kHz **(c)** exactly 20 Hz **(d)** below 20 Hz

9. A car travels 100 km in 2 hours. Its average speed is: 1

(a) 2 km/h **(b)** 200 km/h **(c)** 50 km/h **(d)** 100 km/h

10. A force of 10 N on a mass of 5 kg gives acceleration: 1

(a) 50 m/s squared **(b)** 0.5 m/s squared **(c)** 2 m/s squared **(d)** 15 m/s squared

11. A force of 5 N moves a body 4 m. Work: 1

(a) 20 J **(b)** 0.8 J **(c)** 9 J **(d)** 1.25 J

12. Frequency 100 Hz means: 1

(a) 100 s per oscillation **(b)** 100 m

(c) 100 J **(d)** 100 oscillations per second

13. Displacement is a: 1

(a) unit **(b)** speed **(c)** scalar **(d)** vector

14. The SI unit of force: 1

(a) kg **(b)** metre **(c)** newton **(d)** joule

15. Unit of work: 1

(a) newton **(b)** joule **(c)** watt **(d)** metre

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Define distance. 2

17. State Newton's first law. 2

18. Define work. 2

19. What is sound? 2

20. Define acceleration. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. List three differences between distance and displacement. 3

22. Give three examples of inertia of rest. 3

23. Give three examples of bodies having KE. 3

24. Give three examples of sources of sound. 3

OR

Convert 108 km/h to m/s. 3

25. Find the force on a 1000 kg car with an acceleration of 3 m/s squared. 3

OR

Find work: 20 N through 5 m. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain distance, displacement, speed, velocity and acceleration. 5

OR

State Newton's laws of motion with an example for each. 5

27. Explain work and give the conditions for work to be done. 5

OR

Explain how sound is produced and propagated. 5

END OF QUESTION PAPER — Total: 50 Marks

MINDFIRE